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Noncoding rare variant associations with blood traits in 166,740 UK Biobank genomes

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Diogo M. Ribeiro [1], Robin J. Hofmeister [1], Simone Rubinacci [2,3] & Olivier Delaneau [4]

Large biobanks with whole-genome sequencing (WGS) now enable the association of noncoding rare variants with complex human traits. Given that >98% of the genome is available for exploration, the selection of noncoding variants remains a critical yet unresolved challenge in these analyses. Here we leverage knowledge of blood gene regulation and deleteriousness scores to select noncoding variants pertinent for association with blood-related traits. Integrating WGS and 42 blood cell count and biomarker measurements for 166,740 UK Biobank samples, we perform variant collapsing tests, identifying hundreds of gene–trait associations involving noncoding variants. However, we demonstrate that most of these noncoding rare variant associations (1) reproduce associations known from previous studies and (2) are driven by linkage disequilibrium between nearby common and rare variants. This study underscores the prevailing challenges in rare variant analysis and the need for caution when interpreting noncoding rare variant association results.

Genome-wide association studies (GWASs) have successfully linked common variants to traits, but typically identify variants in linkage disequilibrium (LD) with causal alleles[1]. Although imputation and larger sample sizes now allow exploration of lower-frequency variants[2,3], only whole-genome sequencing (WGS) or whole-exome sequencing (WES) can interrogate the full allele frequency spectrum. Rare variant studies, especially using aggregation tests like SKAT[4,5], offer biological insights complementary to GWASs and may account for some missing heritability[6,7]. The study of rare variation holds great promise, enabling clinical studies in individual patients and is poised to reveal new gene targets for pharmaceutical intervention[8–10]. Notably, the discovery of rare variants in *PCSK9* in individuals displaying low-density lipoprotein cholesterol levels that are low led to clinical trials and the subsequent approval of a PCSK9 inhibitor to treat cardiovascular disease[11,12].

Most rare variant studies focus on coding regions due to the cost-effectiveness of WES[8–10,13], even though exons comprise ~2% of the genome and up to 88% of GWAS associations map to noncoding regions[14,15]. Large-scale noncoding rare variant studies remain limited, primarily due to the need for WGS in large cohorts and challenges in interpreting noncoding effects[16,17]. Recent efforts by biobanks such as UK Biobank[18,19], TopMed[20] and All of Us[21] have begun to address the data gap, but efficient exploration of noncoding regions remains challenging.

Gene expression is regulated by nearby noncoding elements, including enhancers and promoters, typically within 1 Mb of the transcription start site (TSS)[22]. Variants in these regions can influence traits by modulating gene expression[23,24], but mapping regulatory regions to target genes is complex due to incomplete knowledge and cell-type specificity[25,26]. Approaches include chromatin contact mapping (Hi-C)[27,28], correlating enhancer activity with gene expression across individuals[29,30] and various cell types[31,32] and integrated multi-omics models[33,34]. Indeed, genomic data have been used to elucidate the genes and cell types mediating the link between noncoding variants and traits[35,36], a key limitation in GWASs. Translating such approaches to the context of rare variants holds great promise and represents a compelling research avenue.

Here we leverage 166,740 UK Biobank WGS samples to perform genome-wide association analysis of aggregated noncoding rare

[1]Department of Computational Biology, University of Lausanne, Lausanne, Switzerland. [2]Institute for Molecular Medicine Finland (FIMM), University of Helsinki, Helsinki, Finland. [3]Program in Medical and Population Genetics, Broad Institute of MIT and Harvard, Cambridge, MA, USA. [4]Regeneron Genetics Center, Tarrytown, NY, USA. e-mail: diogo.am.ribeiro@gmail.com; olivier.delaneau@regeneron.com

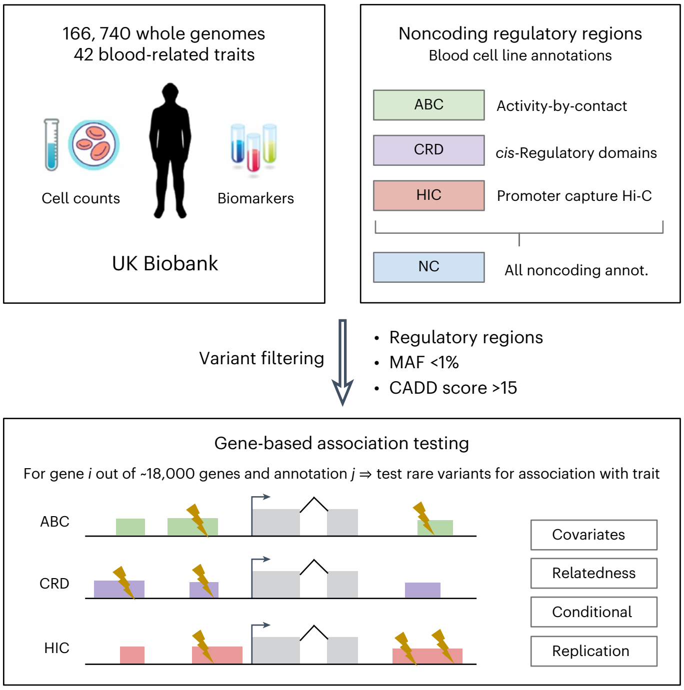

Fig. 1 | Schematic representation of the study approach. An outline of our comprehensive approach to investigating noncoding genetic variants and their associations with 42 blood-related phenotypes in a cohort of white British individuals from the UK Biobank. To explore noncoding regions for potential trait associations, our focus was on gene-regulatory regions derived from three distinct datasets: ABC, CRD and HIC (PCHi-C), spanning various blood cell types. In certain analyses, these three annotations are combined and referred to as the NC (noncoding) annotation. Genetic variants were filtered for MAF < 1% and a CADD score > 15. REGENIE was used for gene-based association tests (SKAT-O tests), including covariates and relatedness estimates. These were performed per gene per trait and per annotation. In addition, follow-up analyses encompassed replication assessments and conditional analyses of GWAS variants.

variants with 42 continuous blood traits, using multiple regulatory annotations from diverse blood cell types. We compare discovery in noncoding versus coding regions and assess independence from known GWAS signals, highlighting the challenge of interpreting noncoding rare variant associations.

Results

Regulatory annotations linking noncoding variants to genes

We evaluated three complementary gene-centric approaches to map noncoding rare variants to protein-coding genes (Fig. 1). First, we gathered *cis*-regulatory domains (CRDs), including 118,408 domains linked to 8,991 genes, inferred from chromatin immunoprecipitation followed by sequencing (ChIP–seq) and RNA sequencing (RNA-seq) in four blood cell types (Table 1). These included histone modifications indicative of active enhancers and promoters (H3K4me1 and H3K27ac) and RNA-seq data obtained from a cohort of individuals, focusing on four key blood cell types (monocytes, neutrophils, T cells and lymphoblastoid cell lines)[29,30]. Second, we obtained 601,733 gene-enhancer links for 13,516 genes from the activity-by-contact (ABC) model, which leverages Hi-C, ChIP–seq and assay for transposase-accessible chromatin with high-throughput sequencing (ATAC–seq) across 60 blood cell types (ABC score > 0.1)[33,34]. Third, we used 808,237 gene–chromatin links to 14,638 genes from promoter capture Hi-C (PCHi-C, hereafter referred to as HIC) data from 17 primary blood cell types[28] (within 1 Mb of TSS, CHiCAGO score > 5). Of note, these annotations differ in their mean proximity to gene TSS, with ABC annotation variants found at closer distances from the gene (absolute median 2.3 kb), whereas CRD and HIC variants are found at larger distances (95.8 kb and 263.6 kb, respectively; Supplementary Fig. 1).

As most genetic variants are expected to have no discernible impact on phenotypes, rare variant association methods benefit from a pre-selection of pertinent variants to include in statistical testing[5,37]. For this, we used combined annotation-dependent depletion (CADD) scores[38], which leverage metrics of noncoding sequence functionality such as sequence conservation, chromatin state and molecular quantitative trait loci to predict pathogenicity for each SNP. We thus considered only rare SNPs (minor allele frequency (MAF) < 1%) with a CADD score > 15, which is equivalent to retaining the ~3.1% most deleterious variants in the genome. We thus retained 2,191,590, 1,062,230 and 11,155,962 autosomal SNPs for the CRD, ABC and HIC annotations, respectively. The variation in SNP numbers across annotations is inherent to the diverse data types, models and cell types used. In addition, the number of SNPs tested per gene is also highly variable, with a median of 243 SNPs in the CRD annotation, 80 for the ABC model and 462 for HIC annotations (Table 1 and Supplementary Fig. 2). By comparison, the coding sequence (CDS) of genes features a median of 165 variants. It is essential to note that regulatory variants overlapping with the CDS regions of any gene were deliberately excluded in noncoding rare variant analysis (Methods).

Noncoding rare variant association to traits

We performed gene-centric SKAT-O tests for each regulatory annotation using REGENIE[39] across 166,740 white British individuals, accounting for relatedness, population structure and covariates (Methods). To understand what can be obtained through rare variant associations, we first present association results without conditioning for GWAS variants. In later sections, we present results conditioning for known GWAS variants. Therefore, we first performed unadjusted association tests against 42 continuous traits, comprising all 23 blood cell counts (for example, monocyte count) and 19 blood biomarkers (for example, creatinine and bilirubin) available in the UK Biobank (Supplementary Table 1). Blood-related phenotypes were deliberately chosen to be aligned with the blood regulatory annotations available, to tackle the issue of tissue and cell-type specificity of gene regulation.

Gene-based association tests, which assess one test per gene (8,991–14,638 genes per annotation), have reduced the multiple-testing burden compared with single-variant GWASs. We therefore considered associations significant at a false discovery rate (FDR) < 5%, corresponding to SKAT-O P values of 6.9×10^{-6} to 6.5×10^{-5}, depending on the annotation. Across all traits, we identified 81–661 significant gene–trait associations, spanning 43–324 genes and 25–42 traits across the 3 regulatory annotations (Table 1, Fig. 2 and Supplementary Table 2). The HIC annotation (highest number of variants) yielded the most associations, whereas ABC (fewest variants) yielded the least. At a stricter threshold ($P < 1 \times 10^{-9}$), we found 11–200 gene–trait associations (4–93 genes, 9–33 traits). Overall, the 3 noncoding annotations were largely orthogonal, with only 9 (0.9%) associations replicated across all 3 and 12–28 across any 2 (Supplementary Fig. 3). All 42 traits had noncoding associations, although some had higher numbers (for example, 59 genes for hemoglobin concentration) than others (for example, 3 for creatinine; Supplementary Fig. 4).

To assess the benefit of combining regulatory annotations, we compared models comprising the union and intersection across the HIC, CRD and ABC annotations (Supplementary Table 3 and Supplementary Fig. 5). The union (NC model) showed >10-fold higher discovery than the intersection and we therefore focused on this model. We then compared noncoding with coding region (CDS) SNPs with a CADD > 15. The NC model identified 911 gene–trait associations versus 723 for CDS (FDR < 5%) and 242 versus 191 at $P < 1 \times 10^{-9}$ (Table 1, Fig. 2 and Supplementary Fig. 6). Noncoding regions also showed more gene–trait peaks, because regulatory SNPs can influence multiple genes[40], unlike CDS SNPs, which typically affect only one gene. For

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Table 1 | Summary of association tests performed across annotations on 42 UK Biobank blood-related traits

Annotation	Genes tested	Median SNPs	Median region size (kb)	Gene–trait FDR<5% before conditioning	Gene–trait $P<1\times10^{-9}$ before conditioning	Gene–trait FDR<5% after conditioning	Gene–trait $P<1\times10^{-9}$ after conditioning
CRD	8,991	243	36.4	278	84	—	—
ABC	13,983	80	19.6	81	11	—	—
HIC	14,638	462	79.4	661	200	—	—
NC	16,673	580	114.9	911	242	65	20
CDS	18,838	165	1.5	723	191	284	78
Promoter	15,080	16	1.0	19	8	8	4

FDR is based on the Benjamini–Hochberg procedure from SKAT-O (two-sided) association *P* values. Only associations significant before conditioning and remaining significant after conditioning at the same threshold are counted.

instance, among 451 NC-associated genes (FDR < 5%), only 171 (38.1%) had no other associated gene within 1 Mb versus 256 of 361 CDS genes (70.1%). In addition to CDS regions, we also identified gene–trait associations from promoter regions, using variants up to 1 kb upstream of the TSS as a proxy (Methods). We found a relatively low number of gene–trait associations, 19 at FDR < 5%, 8 with $P < 1\times10^{-9}$ (Table 1). Promoter regions are smaller than the other noncoding regions used, with a median of only 16 variants per gene.

To assess the utility of using biologically informed annotations, we generated a randomized 'control' annotation. To construct this control set, we first determined the number of variants associated with each gene within the 'NC' annotation. Then, we randomly selected an equivalent number of variants for each gene from a pool of variants adhering to specific criteria: (1) residing within a 1-Mb radius of each gene's TSS, (2) possessing a CADD score > 15 and (3) not being part of either the NC or the CDS annotation (Methods). This provided control variants for ~95% of genes tested in NC. Comparing discovery across 15,687 genes and 42 traits, the control identified 229 and 67 associations at FDR < 5% and $P < 1\times10^{-9}$, respectively (Supplementary Table 4)—a 2.7- to 3.3-fold reduction compared with NC. To confirm our results, we included a further control annotation ('random' annotation) similar to the previous, but without filtering for CADD scores, with the aim of measuring the baseline level of associations from noncoding variants. This additional control showed comparable results to the previous control set, with 351 and 51 gene–trait associations at FDR < 5% and $P < 1\times10^{-9}$, respectively (Supplementary Table 4). Although both control sets underperformed compared with NC, the presence of hundreds of associations suggests that current annotations are incomplete or that additional tissues or cell types are relevant. Moreover, control variants may incidentally lie near causal genes, because the sampling windows often span many genes.

Most rare variant associations to traits are known

We assessed whether the noncoding gene–trait associations discovered here matched discoveries from common variants (GWASs) and coding rare variants (exome-seq). For instance, we identified a noncoding rare variant association between *CST3* (cystatin C) and blood cystatin C levels (Fig. 3a and Supplementary Fig. 7a; SKAT-O $P = 1.7\times10^{-75}$ for CRD annotation). Moreover, we found associations between *IRF8* and monocyte counts (Supplementary Fig. 7b; SKAT-O $P < 5.7\times10^{-61}$ for CRD and HIC annotations), a well-documented association[41,42]. To comprehensively evaluate our ability to identify known gene–trait associations, we gathered gene–trait associations for each of the 42 traits from two datasets: (1) GWAS catalog (common variant analyses) and (2) Genebass databases (coding rare variants from UK Biobank exome-seq data). We found that 21.9% (HIC) and 46.9% (ABC) of gene–trait pairs at FDR < 5% were previously documented in either database, depending on the noncoding annotation (Fig. 3b). This increased to between 24.5% (HIC) and 100% (ABC) when considering gene–trait associations with $P < 1\times10^{-9}$ (Supplementary Fig. 8). This overlap far exceeds the 3.3–4.1% expected by chance, as determined by shuffling SKAT-O *P* values based on the NC annotation. Most CDS gene–trait associations (>90.2%) are known, as expected, because Genebass databases use rare coding variants from the UK Biobank. Indeed, when considering only gene–trait pairs from the GWAS catalog, the replication rate drops to 65.3% (compared with 88.1% in Genebass databases alone, FDR 5%; Supplementary Fig. 9). Intriguingly, use of the control set of noncoding variants (same number of variants as NC) yields similar replication rates as NC (30.7% with FDR < 5%, 34.3% with $P < 1\times10^{-9}$). This suggests that the control variants may reveal bona fide gene–trait relationships with a FDR similar to the regulatory regions—just with a lower number of discoveries, as seen previously.

Given that noncoding variants can influence several nearby genes, we extended our analysis to include nearby genes. Instead of counting only exact gene–trait matches, we included cases where another gene within 1 Mb matched the known gene–trait pair (Methods). This approach found that 84.2% (CRD) to 96.3% (ABC) of gene–trait pairs match, similar to CDS annotations (95.3%, FDR = 5%; Fig. 3c and Supplementary Fig. 8). Although this expanded scan was expected to increase the proportion of matches found, we noted that the expected overlap value when applying random shuffling was 28.0%. These findings underscore that, although noncoding variant analysis presents challenges in pinpointing the exact causal gene compared with coding variants, it effectively identifies likely causal regions, providing valuable insights into the genetic underpinnings of traits.

Rare variant associations overlap GWAS signals

Given that most gene–trait associations identified here are known from previous studies, we next addressed whether these gene–trait associations are independent from known GWAS variants. For this, we used GWAS summary statistics from the Pan-UK Biobank study for the European ancestry population[43]. We derived independent GWAS hits at $P < 1\times10^{-7}$ using the GCTA cojo joint model[44]. Of note, although 98.2% of the GWAS hits are common variants (MAF > 1%), 1,202 conditioning variants are rare (MAF < 1%), 205 overlapping with the tested noncoding variants. Subsequently, we conducted REGENIE gene-based association tests conditioning on all independent GWAS hits per trait (median of 968.5 variants, range 201–2,164). Most gene–trait associations were not independent of GWAS variants, because only 65 out of 911 (7.1%) NC gene–trait associations at FDR < 5% passed significance after conditioning for GWAS variants (Figs. 2e and 4a and Table 1). For associations with $P < 1\times10^{-9}$, 20 out of 242 (8.3%) associations were preserved (Supplementary Table 5). These included well-known associations, such as *CST3* with cystatin C levels (Fig. 3a and Supplementary Fig. 7a; SKAT-O *P* value after conditioning = 1.3×10^{-34}) and several CRISP genes (*CRISP1*, *CRISP2* and *CRISP3*) with reticulocyte phenotypes[45]. Overall, 15% of the 20 associations (compared with 26.7% before conditioning) are replicated in the GWAS catalog or Genebass databases (85% with 'indirect'

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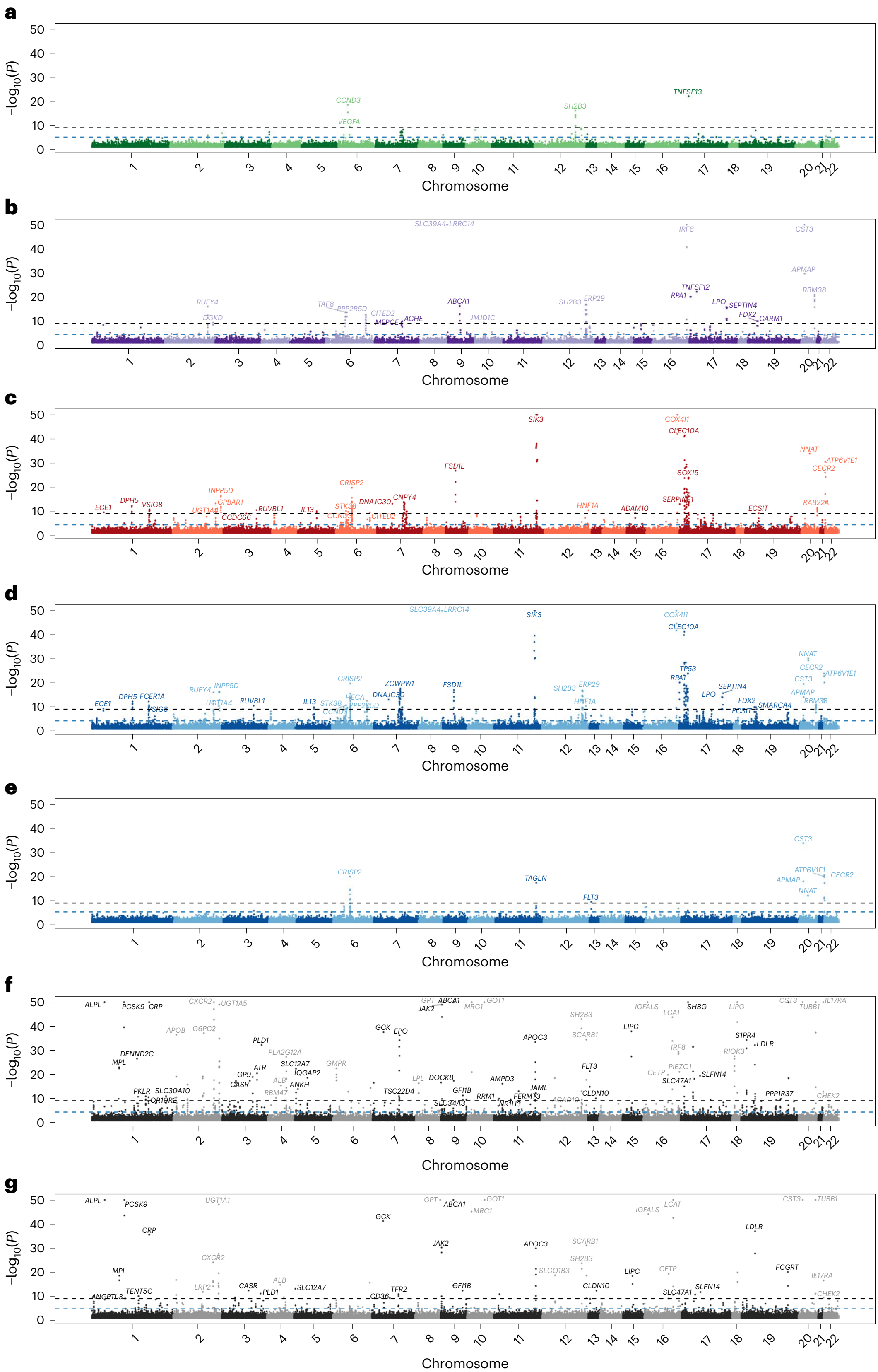

Fig. 2 | Manhattan plot for gene-based rare variant associations across 42 traits and multiple annotations. a, ABC. **b**, CRD. **c**, HIC. **d**, NC. **e**, NC (conditional). **f**, CDS. **g**, CDS (conditional). Each dot represents a gene–trait pair across chromosome positions (x axis, gene TSS position used), with $-\log_{10}(P)$ (capped at 50) of the SKAT-O tests (two sided) on the y axis. **e** and **g** represent association results when conditioning on GWAS variants (see below). The gray lines represent $P < 1 \times 10^{-9}$ cutoff and the blue lines the FDR < 5% cutoff (calculated for each annotation separately through the Benjamini–Hochberg procedure). For visibility, we display only unique gene symbol labels for the most strongly associated gene within 1 Mb passing the $P < 1 \times 10^{-9}$ cutoff.

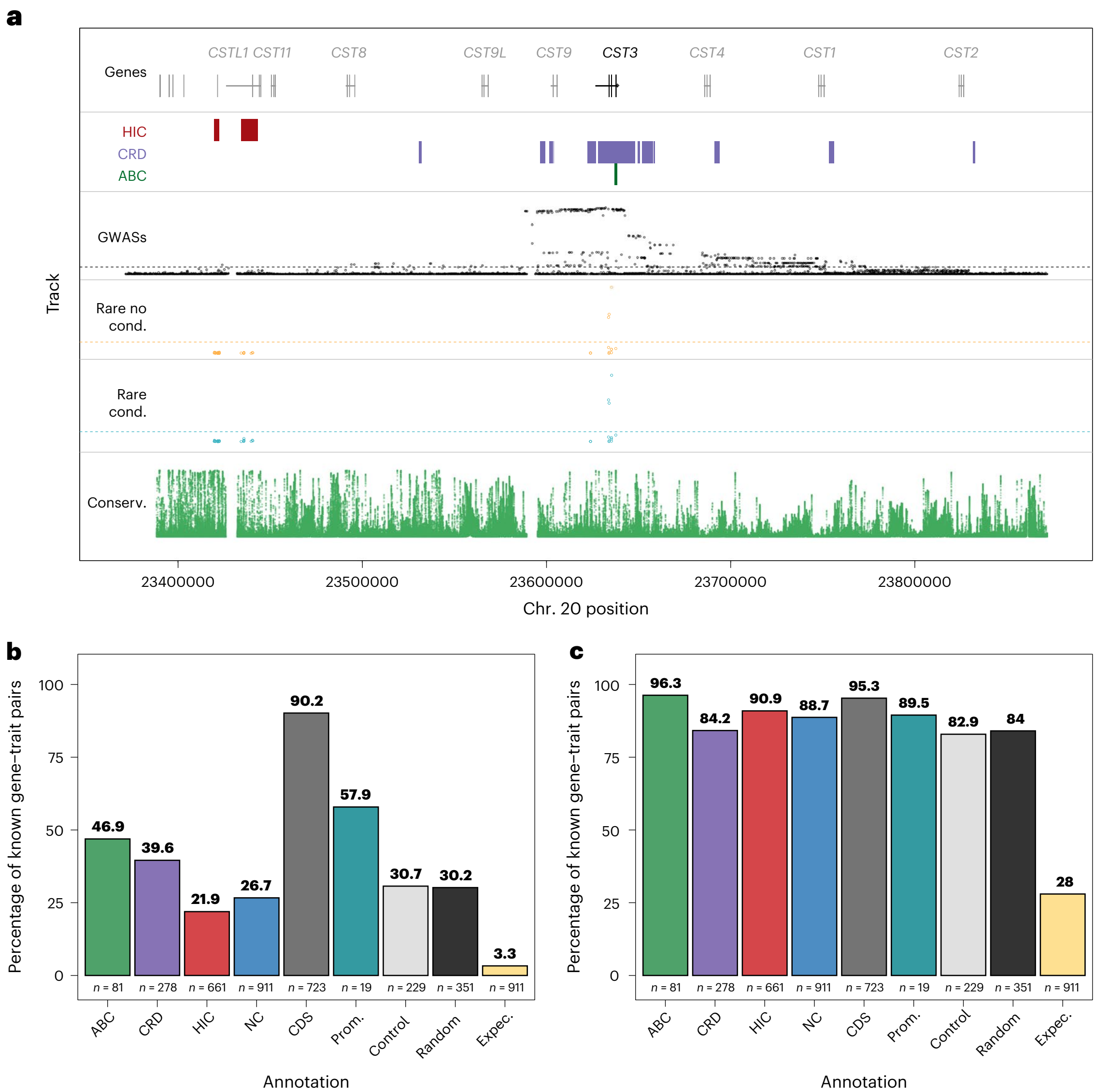

Fig. 3 | Known gene–trait associations. a, Details of *CST3* association with cystatin C levels. The region focuses on 250 kb around the gene TSS that includes tested rare variants. *CST3* is represented in black and other nearby genes in gray. HIC regions tested for rare variants are represented in red rectangles, CRD in purple and ABC in green. Pan-UKB GWAS summary statistics are shown as black dots ($-\log_{10}$ scaled from 0 to 1); single rare variant association tests before conditioning on GWAS variants are shown in yellow ($-\log_{10}$ scaled from 0 to 1); single rare variant association tests after conditioning (cond.) are shown in cyan ($-\log_{10}$ scaled from 0 to 1); and conservation (Conserv.) metrics from PhastCons alignments of 30 mammals (27 primates) are shown in green. Dashed lines represent $P = 1 \times 10^{-9}$ significance threshold. **b**, Percentage of gene–trait pairs at FDR = 5% significance present on GWAS catalog or Genebass databases depending on the annotation. **c**, Percentage of gene–trait pairs at FDR = 5% present on GWAS catalog or Genebass databases through either direct gene–trait associations or indirect associations within a 1-Mb window. expec., expected (random shuffling of NC annotation association *P* values); prom., promoter.

replication; Supplementary Fig. 10). Besides being independent of GWAS hits, we found that all 20 noncoding associations ($P < 1 \times 10^{-9}$) also remained significant after conditioning for all tested CDS variants in the respective gene (Supplementary Fig. 11), suggesting that the association mechanism is regulatory. For comparison, conditioning on CDS gene–trait associations retained between 39.2% and 40.8% associations at FDR < 5% and $P < 1 \times 10^{-9}$, respectively (Figs. 2g and 4b), a much higher proportion than NC associations. Similarly, although promoter regions identified a low number of associations overall (8–19), 42–50% of those associations remained significant after conditioning for GWAS variants (Table 1). In control and random regions, only a few associations remained significant after conditioning (2.2% at FDR < 5%, 1.5% at $P < 1 \times 10^{-9}$ for control, 0% for random; Supplementary Table 4), highlighting the importance of biologically relevant regions.

To explore whether LD explains the drop in significance when conditioning for GWAS variants, we focused on associations driven by a single variant. We thus performed single-variant analysis for variants in NC and CDS annotations before conditioning for GWAS variants (Methods). Then, we categorized significant gene–trait associations from the collapsing analysis into: (1) cases where single variant(s) attain a lower *P* value than the SKAT-O test, suggesting that the collapsing signal is driven by single variants, and (2) cases where no single variant exhibited a lower *P* value than the SKAT-O test, indicating that only a collapsing test could unveil the association. Furthermore, we moved cases 1 ('single cases') to 2 ('group cases') when SKAT-O tests remained significant after conditioning for the single variants in case 1, which indicated that the collapsing signal was independent of the significant single variants (Supplementary Fig. 12 and Methods). We found that 66% ($P < 1 \times 10^{-9}$) to 70% (FDR < 5%) of NC gene–trait associations could

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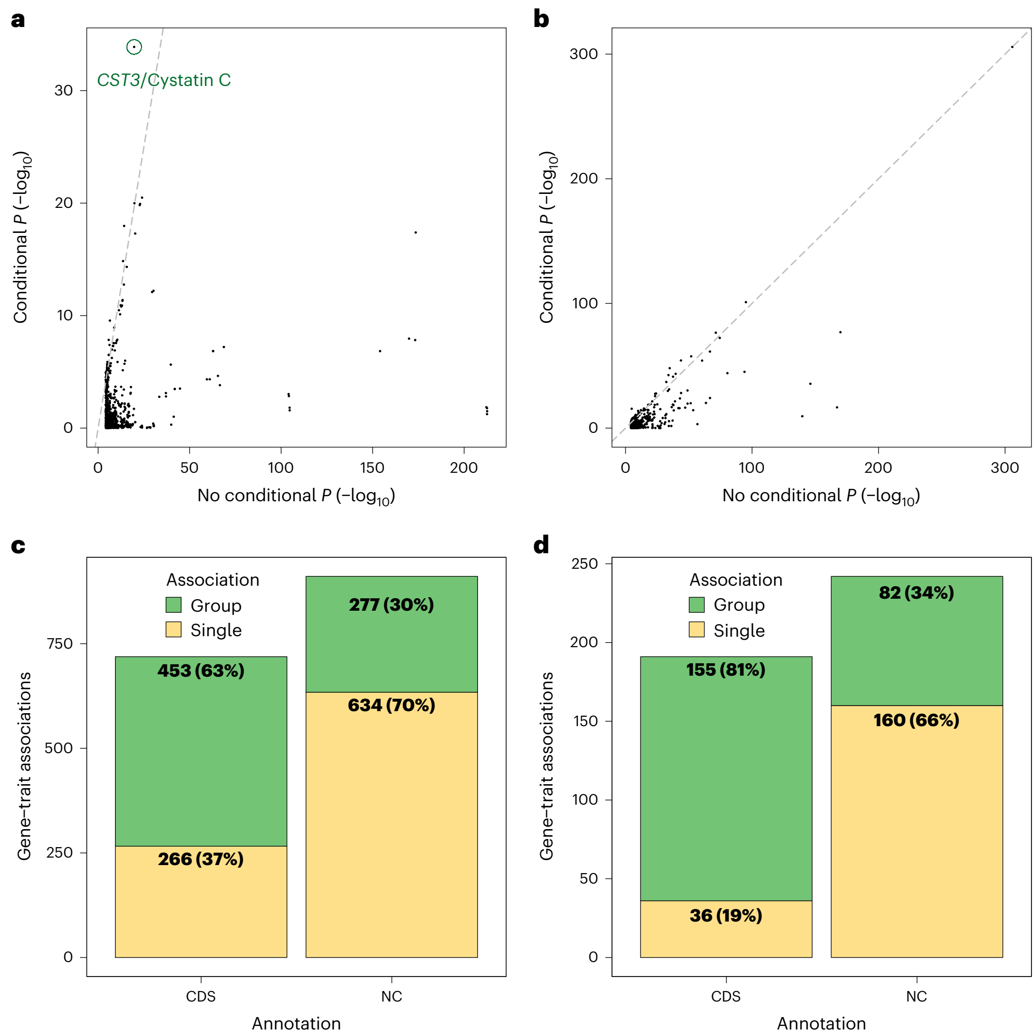

Fig. 4 | Rare variant association type and conditioning analysis. a, NC gene–trait associations (911 associations) $-\log_{10}$(SKAT-O) (two-sided) *P*-value comparison before and after conditioning on GWAS variants per trait. **b**, As for **a** but for the 723 CDS associations. Note the difference in the *y* axis range compared with the previous plot. **c**, Categorization of FDR < 5% (Benjamini–Hochberg procedure) CDS and NC gene–trait associations into single and group. **d**, Categorization of gene–trait associations with $P < 1 \times 10^{-9}$.

be attributed to a single variant (case 1; Fig. 4c,d, Supplementary Fig. 12 and Supplementary Table 6), whereas 30–34% were identifiable only through collapsing tests (case 2; Supplementary Table 7). Notably, these patterns differed for CDS annotations, where a higher proportion of associations could be identified only through collapsing tests (63% at FDR < 5%, 81% at $P < 1 \times 10^{-9}$), with only 19–37% CDS associations driven by a single variant (Fig. 4c,d). After conditioning for GWAS variants, a comparable proportion of group and single cases was found (87% group for CDS, 18% for NC, FDR = 5%; Supplementary Fig. 13).

We next examined the effect of LD between GWAS variants and rare variants, focusing on 153 rare single variants associated in the 634 NC single cases (FDR < 5%; Fig. 4c). For each single variant and associated trait, we gathered GWAS conditional variants within 1 Mb, yielding 114 single variants with a match and 695 'single-conditional' variant pairs. We then calculated the LD between each pair of variants. We found that 99.1% (113 out of 114) single variants were in LD with at least one GWAS conditional variant, as measured by the *D* prime (*D*′) LD metric (*D*′ > 0.5; Supplementary Table 8). Indeed, conditioning on GWAS variants in LD reduced significance for 74.4% of variant–trait associations (206 associations; Supplementary Fig. 14 and Supplementary Table 9). Specifically, 38.3% (79 out of 206) variant–trait associations with FDR < 5% and 43.6% with $P < 1 \times 10^{-9}$ no longer passed this threshold after conditioning, compared with 69.4% and 78.2% when conditioning for all GWAS variants across chromosomes. It is interesting that we found that variants in associations that remained significant (127 cases out of 206) have lower allele frequencies (mean = 0.0038) compared with variants in associations that lose significance (79 cases out of 206, mean = 0.0063, Wilcoxon's rank-sum test $P = 6.4 \times 10^{-8}$; Supplementary Fig. 14). Accordingly, variants that remain significant also displayed lower LD (mean = 0.03) compared with those losing significance (mean = 0.25, Wilcoxon's rank-sum test, $P = 4.6 \times 10^{-19}$).

To further investigate whether the results obtained are driven by low-frequency or rarer variants, we identified gene–trait associations for the NC annotation including only variants with MAF < 0.1% in SKAT-O tests. Before conditioning, 11–19× more associations were found with MAF < 1% (911 associations, FDR < 5%) than with MAF < 0.1% (47 associations; Table 2 and Supplementary Table 10). However, after conditioning, the number of gene–trait associations is similar, with 40 associations at FDR < 5% identified with MAF < 0.1% (compared with 65 for MAF < 1%) and 18 remaining at $P < 1 \times 10^{-9}$ (compared with 20 for MAF < 1%; Supplementary Table 11). In fact, conditioning for GWAS variants on MAF < 0.1% has visibly less impact on the *P* values than for MAF < 1% (Fig. 4a and Supplementary Fig. 15). As expected, there is a strong overlap between gene–trait pairs identified with MAF < 1% and

Table 2 | Summary of NC annotation association tests between MAF <1% and MAF <0.1% rare variant thresholds

Annotation	Genes tested	Median SNPs	Gene–trait FDR <5% before conditioning	Gene–trait $P<1\times10^{-9}$ before conditioning	Gene–trait FDR <5% after conditioning	Gene–trait $P<1\times10^{-9}$ after conditioning
MAF <1%	16,673	580	911	242	65	20
MAF <0.1%	16,626	554	47	22	40	18

The results of SKAT-O tests (two sided) before and after conditioning for GWAS variants are shown. The FDR is based on the Benjamini–Hochberg procedure.

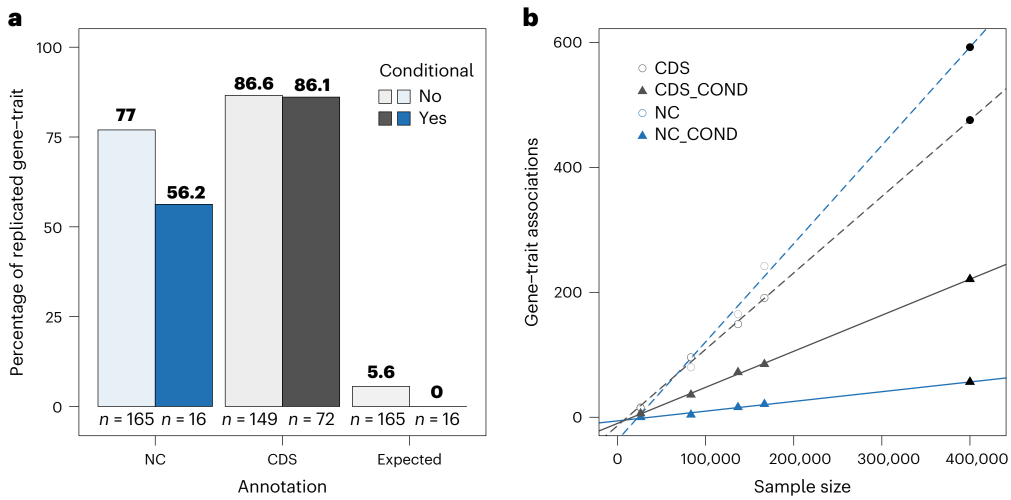

Fig. 5 | Replication of discovered gene–trait associations. a, Percentage of gene–trait SKAT-O (two-sided) associations with $P<1\times10^{-9}$ discovered in a set of 136,740 samples that have association at $P<0.05$ in 26,212 unrelated samples, before and after conditioning for GWAS variants. 'Expected' refers to a random shuffling of NC gene–trait association P values. **b**, Correspondence between sample size and number of discovered gene–trait associations at $P<1\times10^{-9}$. Values at 400,000 samples are a prediction based on the previous four sample sizes tested.

MAF < 0.1%, with three associations found exclusively with MAF < 1% and $P<1\times10^{-9}$ and only one found exclusively with MAF < 0.1% (*IGFBP3* and insulin-like growth factor 1 levels; Supplementary Fig. 16). This suggests that variants with <0.1% MAF may drive many associations identified with the MAF < 1% threshold, because they are less likely to be in LD with GWAS variants. Yet, using variants with MAF < 1% finds several additional associations compared with MAF < 0.1% (31 at FDR < 5%, 3 at $P<1\times10^{-9}$), with minimal loss (5 at FDR < 5% and 1 at $P<1\times10^{-9}$), allowing for a more comprehensive analysis.

Out-of-sample replication and sample size

Next, we evaluated the replication rate of gene–trait associations across different samples. For this, we randomly split the set of 166,740 WGS samples into discovery (136,740 samples) and replication (26,212 samples), excluding related samples between the groups. We found that 77.0% of 165 NC gene–trait associations ($P<1\times10^{-9}$) before conditioning were replicated at $P<0.05$, compared with 5.6% by chance (Fig. 5a). After conditioning for GWAS variants, 56.2% of 16 associations replicated (none expected by chance). In comparison, CDS regions had higher replication rates, with 86.6% before and 86.1% after conditioning. Moreover, association P values before conditioning were moderately correlated between discovery and replication sets (Supplementary Fig. 17; Spearman's correlation = 0.39, $P<2.2\times10^{-16}$).

Sample size had a strong effect on discovery. Downsampling to 26,212, 83,370, 136,740 and 166,740 samples showed a strong linear relationship with the number of associations discovered (NC correlation = 0.98, $P=0.013$; CDS correlation = 0.99, $P=2.7\times10^{-3}$; Fig. 5b). Comparing 166,740 with 83,370 samples (twofold difference), NC associations post-conditioning increased fourfold (16 versus 4 at $P<1\times10^{-9}$). These results indicate that discovery is far from plateauing. Indeed, projecting to 400,000 samples (approximate size of the full UK Biobank for British samples), we would expect 56 NC and 221 CDS associations independent of GWAS variants at $P<1\times10^{-9}$.

Finally, given that some of the traits explored in our analysis may be regulated in tissues other than blood (for example, creatinine, albumin and bilirubin), we identified gene–trait associations for EpiMap[32] gene-enhancer annotations for liver, kidney, pancreas and endothelial cells, as well as control T cells and B cells. We identified 25 gene–trait associations at $P<1\times10^{-9}$ across the 6 annotations, after conditioning for GWAS variants (Table 3, Supplementary Fig. 18 and Supplementary Table 12). It is interesting that 15 of these gene–trait associations are not significant at $P<1\times10^{-9}$ in the NC annotation, with 8 specific to non-blood tissues (Supplementary Fig. 19). However, most of these associations are reported in the GWAS catalog or Genebass databases (Supplementary Fig. 20). We found lower numbers of gene–trait associations with EpiMap blood annotations (T cell and B cells) than NC annotations (Table 3). This likely reflects fewer variants per gene (median 21–31 versus 580 for NC) and fewer genes tested (11,777 versus 16,673) in EpiMap annotations than NC annotations. Overall, these results suggest that the exploration of multiple tissues and cell types may reveal further associations, yet more comprehensive gene regulation annotations for non-blood tissues and cell types are required to fully assess their contribution.

Discussion

Expression quantitative trait loci have been used to aid interpretation of GWAS associations, but often account for only a small fraction of GWAS signals[46,47]. Given these shortcomings, an alternative is to use genomic data on regulatory regions to identify genes mediating the impact of noncoding variation on phenotype[34,35]. In this study, we used biobank-scale WGS data and regulatory genomics to address the challenge of linking noncoding rare variants to human traits.

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Table 3 | Summary of association tests for additional tissues and cell types from the EpiMap resource

Annotation	Genes tested	Median SNPs	Gene–trait $P<1\times10^{-9}$ before conditioning	Gene–trait $P<1\times10^{-9}$ after conditioning
Liver	12,846	28	25	9
Kidney	12,035	31	14	6
Pancreas	10,845	22	8	1
Endothelial cells	10,436	23	9	0
T cells	11,673	21	19	3
B cells	12,827	28	21	6

Number of significant SKAT-O test (two-sided) gene–trait associations with $P<1\times10^{-9}$ before and after conditioning for GWAS variants across 42 blood-related traits.

Previous studies have associated noncoding rare variants using sliding windows[48,49], functional annotations (for example, enhancers)[50–52] and variant deleteriousness regions[17,37], but these were typically limited to <10 traits or smaller cohorts (<35,000 samples). Here we combined variant deleteriousness filtering with comprehensive gene regulation datasets to study associations between noncoding rare variants and dozens of traits in >166,000 individuals. Due to the lack of consensus on noncoding region annotation, we used multiple gene-regulatory annotations to narrow the vast search space inherent to the study of noncoding regions. In addition, we aligned regulatory annotations with the specific cell types relevant to the traits under scrutiny, with the intuition that the use of biologically relevant annotations is better suited to map causal loci. This pilot focused on blood traits and related annotations, given their rich data and broad disease relevance. We noted that more genomic data are available for white blood cells than for red blood cell and platelet lineages. Although we included megakaryocytes and erythroblasts[28,34] and identified many genes linked to red blood cell and platelet traits, some associations may still be underestimated[45]. Moreover, several biomarkers (for example, creatinine and sex hormones) are also regulated outside blood. Although we evaluated other tissues (liver, kidney, pancreas and endothelial cells), the breadth of data available is reduced compared with blood, resulting in reduced discovery power. Nevertheless, our approach is generalizable to any cell type or trait, contingent on data availability.

Our approach identified 911 noncoding gene–trait associations across 42 traits before conditioning on GWAS variants—more than the 723 associations found in CDSs using the same approach. We, however, would like to point out that CDS discovery was not optimized, because we did not use coding annotations like missense or loss-of-function variants, which are known to enhance detection[10,13]. Moreover, although noncoding regions yielded more associations, interpretation proved more complex. Coding variants typically affect one gene, but regulatory variants (for example, in enhancers) may influence multiple nearby genes[24,40]. Our method accounted for this by allowing multiple gene associations per variant, increasing the total number of gene–trait associations but complicating identification of causal genes. Roughly 28–54% of noncoding associations overlapped with entries in the GWAS catalog or Genebass databases, but this rose to nearly 100% when including genes within 1 Mb. This indicates that our method highlights potentially causal regions and gene sets, but often lacks gene-level specificity. Although nearly all our discovered gene–trait associations may already be documented, it should be noted that the GWAS catalog and Genebass databases include SNP array and exome data from the UK Biobank. Although large WGS datasets are now emerging, our findings do not show a higher discovery yield from using WGS—aligning with recent work suggesting limited benefit of WGS over SNP arrays, imputation and exome sequencing within the UK Biobank[53].

With growing sample sizes and improved imputation, modern GWASs now examine variants with MAF between 0.1% and 1% (refs. 54,55). A study of 5 million individuals found that both common and low-frequency variants are associated with height in the same loci[54]. Consistent with this, we observed that many of our rare variant associations are not independent of known GWAS signals. Indeed, we estimated that a large fraction of associated rare variants are in LD with previously identified GWAS variants and accounting for those GWAS variants often decreases rare variant signals. Although a straightforward interpretation would be that common variants are causal and we find rare variants associations only due to LD[51,52], rare variants are by definition underpowered in association testing compared with common variants. This makes it difficult to determine whether the signal comes from the common or rare variants. Indeed, it has been proposed that common variant associations from GWASs could arise from 'synthetic associations' due to nearby rare variants[56], although it is unclear how often this occurs[57,58].

Although this study represents a major scale-up, larger datasets like the UK Biobank's upcoming release of 500,000 genomes will further boost power. We estimated that discovery power with both coding and noncoding rare variant analysis will clearly benefit from increased sample size. In addition, the incorporation of phasing information may improve association modeling by identifying whether one or both alleles are affected[59]. Finally, variant selection is key in rare variant studies; we used CADD scores to prioritize deleterious variants[17,37,60], enabling consistent comparisons across coding and noncoding regions. However, exploration of alternative scores—ReMM, GERP, Eigen[61–63] for noncoding regions and AlphaMissense[64] for coding regions—constitutes an open avenue of research.

In summary, our findings suggest that most rare noncoding associations reflect known signals and LD between common and rare variants. This highlights the need for caution in interpreting rare variant associations, which should always be supported by robust conditional analyses. Further research is needed to clarify the role of noncoding rare variants in identifying independent trait associations, including studies on additional traits and larger sample sizes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-025-02288-x.

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Methods

Ethics statement

This study relied on analyses of genetic data from the UK Biobank (UKBB) cohort, which was collected with informed consent obtained from all participants. Data for this study were obtained under the UKBB application license no. 66995. All data used in this research are publicly available to registered researchers through the UKBB data-access protocol.

Noncoding rare variant annotation

We considered three orthogonal noncoding rare variant annotations in blood cell types described below.

HIC. This dataset consists of HIC (promoter capture Hi-C data (PCHi-C)) data from 17 human primary hematopoietic cell types from ref. 28. We considered only intrachromosomal regions within 1 Mb of the promoter region of a gene. Only HIC interactions with CHiCAGO scores > 5 in at least one cell type were used.

CRDs. We obtained *cis*-regulatory domains (CRDs) derived from correlation between histone modification ChIP–seq peaks (H3K4me1 and H3K27ac) across 94–317 individuals and 4 blood cell types from ref. 30 (neutrophils, monocytes and T cells) and ref. 29 (lymphoblastoid cell lines). RNA-seq on the same samples was used in the original studies to link CRDs to genes. We combined CRD–gene links across the four cell types with bedtools merge -o distinct (v.2.26) to remove redundant regions per gene.

ABC. We obtained gene-enhancer association predictions from ref. 34 (AllPredictions.AvgHiC.ABC0.015.minus150.ForABCPaperV3.txt.gz). We considered only interactions from a set of 60 blood cell types. Moreover, as done by the data providers, we analyzed only interactions with ABC score > 0.1.

We further combined these three regulatory annotations into a single NC annotation, comprising any variant falling in any of the three annotations above. For all noncoding annotations, to ensure that regulatory association signals were not driven by coding regions, we excluded any overlap with CDS regions (as defined by Gencode v.42) using bedtools subtract (v.2.26). We used these same CDS regions as a comparison for association testing. We further defined promoter regions by including variants within 1 kb upstream of the gene TSS region (as defined by Gencode v.42). We noted that, although by this definition all genes have a promoter region, not all genes have at least one variant in this relatively small region. In all analyses, we considered only autosomal protein-coding genes based on Gencode v.42 and excluded genes in the major histocompatibility complex region (chr6:28,510,120-33,480,577). When required, we lifted coordinates from hg19 to GRCh38 using the UCSC (University of California, Santa Cruz) liftOver tool[65] and converted gene symbols to Ensembl IDs using gProfiler (v.105)[66].

We produced two control annotations (control and random) to compare with the NC annotation. For this, for each gene we randomly selected SNPs matching the number used in the NC annotation, considering a 1-Mb window around the gene TSS, while excluding coding regions as well as any variant used in the NC annotation. For the control annotation, we further excluded variants with a CADD score < 15, to match the approach undertaken for the NC annotation. After these filters, around 5% of the genes did not have enough remaining variants to match the number of variants in the NC annotation. We excluded these genes when calculating association FDR and comparing numbers of NC, control and random gene–trait associations.

We obtained gene-enhancer links for additional tissues and cell types (liver, kidney, pancreas, endothelial cells, B cells and T cells) from the EpiMap[32] resource 'links by group' files (https://personal.broadinstitute.org/cboix/epimap/links/pergroup). We lifted hg19 coordinates to GRCh38 coordinates using the UCSC liftOver tool. We processed these annotations as before, excluding nonprotein-coding genes, CDS regions and major histocompatibility complex genes, as well as applying a CADD > 15 filtering on SNPs.

UKBB WGS and phenotypes

We used the 200K WGS dataset release from the UKBB, processed as described in the SHAPEIT Phasing Protocol for 200K WGS (field 20279). Briefly, starting with GraphTyper pVCFs, we excluded variants with (1) allele assignment (AA) score < 0.8, (2) Hardy–Weinberg equilibrium test $P < 1 \times 10^{-30}$ (calculated in unrelated European ancestry samples), (3) genotype missingness >10% and (4) excluded variant without a 'PASS' in the VCF file. Multiallelic variants were converted to biallelic sites using bcftools norm (with -m -any parameters, v.1.15) before association testing.

We considered only 166,740 individuals of white British European ancestry (data field 22006, that is, excluded 33,111 individuals due to ancestry) who met the following criteria: (1) reported and genetic sex consistent (fields 31 and 22001); (2) fewer than 10 third-degree relatives within the dataset (field 22021); and (3) no outliers in terms of heterozygosity or missing data rates (field 22027). We considered related samples, because these relationships are accounted for in association testing (see below).

We collected phenotypic data for a total of 42 blood-related traits, including 23 measures of blood cell counts and percentages, as well as 19 blood biomarkers (mean = 157,149 samples; Supplementary Table 1). In cases where multiple measurements of the same phenotype were available for a single individual, the first instance was used. We also collected age, sex and genotype principal components for each individual.

Rare variant association testing

Variant selection. Before association testing, we filtered for UKBB WGS variants falling in any regulatory annotation position or CDS region with bcftools (v.1.10.2). To test only variants predicted to be highly deleterious, we retained only variants with a CADD (v.1.6) Phred-scaled quality score > 15 ('All possible SNVs of hg38')[38] using customized R (v.4.0.1) and Python (v.3.7.2) scripts. Of note, only CADD scores for single nucleotide variants (SNVs) were used and, thus, only this type of variant was assessed in this study. VCF files were converted to PGEN using plink2 (--make-pgen erase-phase) for association testing.

Gene-based association testing. We performed genome-wide association tests using REGENIE (v.3.2.8)[39] for 166,740 individuals of white British European ancestry with WGS data. For REGENIE step 1, we included 360,166 UKBB SNP array (field 22418) variants for the same individuals, considering only variants in autosomes with MAF > 1%, <10% genotype missingness, Hardy–Weinberg equilibrium test $P > 1 \times 10^{-15}$ and pruned for LD with plink2 (parameter --indep-pairwise 500 50 0.2). The leave-one-chromosome-out models obtained with step 1 of REGENIE are used for step 2, for each trait. In step 2, we used sex (field 31), age (field 21022) and 10 ancestry-informative principal components derived from SNP array data (field 22009) as covariates. REGENIE step 2 for gene-based tests was run with the following parameters: --minMAC 1, --bsize 400, --aaf-bins 0.01, --vcf-tests skato, --vc-maxAAF 0.01, --apply-rint (that is, rank-inverse normalized phenotype). Of note, this version of REGENIE collapses ultra-rare variants (defined as minor allele count <10) into a single unit. We ran REGENIE step 2 separately for each annotation. Note that the output of SKAT-O tests does not include estimates of β (effect size) and its s.e. and, thus, these are not reported in this study. We calculated the FDR (Benjamini–Hochberg procedure) separately for each regulatory annotation, because the NC annotation comprises a single mask including variants from ABC, CRD and HIC annotations. We then considered two significance thresholds for gene–trait associations: FDR < 5% and a more stringent $P < 1 \times 10^{-9}$.

Replication. To assess replication of association in different UKBB samples, we split the 166,740 samples into 30,000 samples chosen randomly and the remaining 136,740 samples. From the 30,000 samples, we excluded those with at least 1 relative in the remaining set of 136,740 samples, retaining 26,212 samples. We then performed association separately on the set of 26,212 samples and the remaining 136,740 samples. In addition to these subsets of samples, gene–trait associations were also computed for a random subset of 83,370 samples (that is, half of the 166,740 samples) to study the relationship between sample size and discovery power.

Single-variant association testing. We performed association tests per variant against all 42 traits using REGENIE, from the same sets of variants used in gene-based tests. We ran step 2 with the same parameters (except --vcf-tests skato), covariates and step 1 models as for gene-based tests. We included all single variants with MAF < 1%. Next, we compared association *P* values between gene-based and single-variant tests to determine which associations are driven by one or more combinations of variants. For this, for each significant gene–trait association from gene-based tests (NC and CDS annotations), we determined whether at least one single variant linked to the same gene had a *P* value as significant or more significant than the gene-based test (SKAT-O *P* value). These cases were labeled as single cases. For the remaining gene–trait associations (that is, cases where gene-based association had higher significance than any single variant linked to the gene), we classified these cases as group cases. Finally, we moved single cases to group cases if the SKAT-O collapsing test remained significant (at FDR 5% or $P < 1 \times 10^{-9}$) after performing additional tests conditioning for single variant(s) with a lower *P* value than the SKAT-O test (median 1 variant per gene–trait pair, maximum 4 single variants). This classification was performed both before and after conditioning for GWAS variants.

Overlap with known associations

We evaluated how many of our gene–trait association predictions overlap known gene–trait associations from two sources: (1) GWAS catalog[67] (All associations v.1.0, downloaded in March 2023), keeping all genes reported by study authors ('Reported gene(s)' column) as well as genes overlapping or upstream or downstream of an associated SNP ('Mapped gene(s)' column); and (2) Genebass databases[13] based on UKBB exome-seq data. We obtained Genebass associations through Hail in August 2022 ('results.mt') and considered SKAT-O *P* values <5% FDR per trait for the 'pLoF' and 'Missense|LC' categories. All 42 traits had a match in both databases. A gene–trait pair was deemed replicated if the pair was observed in at least one of the databases. The 'Expec.' results were computed by shuffling, once, the SKAT-O *P* values from the NC annotation, before FDR calculation and thus drawing random sets of gene–trait pairs matching the same number of discoveries. We used this to measure the number of gene–trait associations across traits expected to be reproduced by chance.

To study groups of genes in a locus, we also measured the overlap between associated and known gene–traits when a gene within 500 kb of each TSS region flank matched. For this, for each gene–trait pair, we gathered all genes within this flanking region. If any of these genes had a known association to the same trait, we counted it as a match. If neither the current gene nor nearby genes matched, a match was not counted.

Association conditional analysis

To condition for known common variants associated with traits, we first obtained GWAS summary statistics for the EUR population from the Pan-UKBB[43]. These GWAS computations encompassed the samples used in our study, as well as others with SNP array information (mean = 369,748 samples for the traits assessed). We then obtained independently associated common variants for each trait by using the GCTA (v.1.94.1) cojo joint model function[44], with the following parameters: window = 10 Mb, collinearity = 0.9, for all GWAS variants with $P < 1 \times 10^{-7}$. To run GCTA cojo models, we used LD structure estimates for GWAS common variants (WTCHG-imputed SNP array data, field 2282) from 10,000 UKBB samples randomly sampled from the set of white British individuals with WGS data. We used variants with $P < 1 \times 10^{-7}$ (instead of the typically used $P < 5 \times 10^{-8}$ cutoff) to ensure that all known common variant signals are accounted for, as done in previous rare variant studies[10]. We performed conditional analysis for genotypes of independent GWAS hits (SNP array data, field 2282) using REGENIE --condition-list and --condition-file. Unless stated otherwise, we used all independent GWAS hits across all chromosomes associated with a trait together when conditioning. We kept all other parameters and REGENIE step 1 models as in the analysis without conditioning. To estimate independence of NC gene–trait associations from CDS variants in the same gene, we performed REGENIE SKAT-O conditional tests for the 20 NC associations with $P < 1 \times 10^{-9}$, by accounting for all CDS variants with CADD > 15 (that is, based on the CDS annotation used previously; median of 65 variants per gene) in addition to the GWAS variants used before.

We calculated LD r^2 and D' metrics using the plink2 --ld flag. First, we lifted SNP array hg19 coordinates to the hg38 assembly with the UCSC liftOver tool. Then, for each of 245 rare variants associated with a trait at FDR < 5%, we gathered all independent GWAS common variants (from GCTA cojo) falling within a 1-Mb window. A total of 163 unique rare variants had at least 1 common variant within this window. We measured LD across 1,006 rare and common variants paired in this manner.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The lists of gene–trait associations identified here are available as Supplementary Tables 1–12. These lists and other source data are available via Zenodo at https://doi.org/10.5281/zenodo.15546894 (ref. 68). The WGS data and individual phenotypes used for association testing can be accessed via the UKBB Research Analysis Platform: https://ukbiobank.dnanexus.com/landing. This platform is open to researchers who are listed as collaborators on UKBB-approved access applications. Public regulatory annotation datasets used include: the ABC models, available at https://www.engreitzlab.org/resources; promoter capture Hi-C data, obtained from ref. 28, accessible at https://osf.io/u8tzp; and CRDs obtained from the original publications[29,30]. CADD scores of deleteriousness are available at https://cadd.gs.washington.edu. Known gene–trait associations used are available in the GWAS catalog (https://www.ebi.ac.uk/gwas) and Genebass databases (https://app.genebass.org). Source data are provided with this paper.

Code availability

Code to reproduce analysis and plots are avilable via GitHub at https://github.com/diogomribeiro/noncoding_rarevariant and via Zenodo at https://doi.org/10.5281/zenodo.15546894.

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Author contributions
D.M.R. and R.J.H. performed the experiments. D.M.R. analyzed the data and wrote the manuscript with input from O.D. D.M.R., S.R. and O.D. conceived, designed and managed the study.

Competing interests
O.D. is a current employee of Regeneron Genetics Center, which is a subsidiary of Regeneron Pharmaceuticals. D.M.R. currently works as a contractor for F. Hoffmann-La Roche AG. The other authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41588-025-02288-x.

Correspondence and requests for materials should be addressed to Diogo M. Ribeiro or Olivier Delaneau.

Peer review information *Nature Genetics* thanks Andrew Johnson, Jacob Ulirsch and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Corresponding author(s): Olivier Delaneau
Diogo Ribeiro

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Reporting Summary

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☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
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☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for the data collection of this study.
Data analysis	Third party code and programs used in this study all are publicly available: bcftools (vl.10.2, v1.15) bedtools v2.26 PUNK v2.0 REGENIE v3.2.8 GCTA vl.94.1 R v4.0.1 Python v3.7.2 gProfiler v105 Custom code developed as part of the study: R and python scripts: https://github.com/diogomribeiro/noncoding_rvat

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The lists of gene-trait associations identified here are available as Supplementary Tables. The WGS data and individual phenotypes used for association testing can be accessed via the UKB Research Analysis Platform (RAP): https://ukbiobank.dnanexus.com/landing. This platform is open to researchers who are listed as collaborators on UKB-approved access applications. Public regulatory annotation datasets used include: the Activity-by-Contact models, available at https://www.engreitzlab.org/resources; promoter capture Hi-C data, obtained from Javierre et al. 2016, accessible at https://osf.io/u8tzp/; cis-regulatory domains (CRDs), obtained from the original publications by Delaneau et al. 2019 and Avalos et al. 2023. CADD scores of deleteriousness are available at https://cadd.gs.washington.edu/. Known gene-trait associations used are available at the GWAS catalog (https://www.ebi.ac.uk/gwas/) and Genebass (https://app.genebass.org/).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	No sex or gender-based analysis were conducted, we only produced results for autosomes.
Reporting on race, ethnicity, or other socially relevant groupings	No race, ethnicity or other social grouping results were reported. Only one population group was analysed, defined by genetic principal components analysis (produced and defined by the data providers, UK Biobank).
Population characteristics	Participants were 166,740 UK Biobank individuals of white British ancestry, aged 40–69 years at recruitment (mean ~56.5 years), including both sexes (approximately 54% female). Analyses adjusted for age and sex.
Recruitment	N/A (performed by the UK Biobank)
Ethics oversight	This study relied on analyses of genetic data from the UKB cohort, which was collected with informed consent obtained from all participants. Data for this study were obtained under the UKB application licence number 66995. All data used in this research are publicly available to registered researchers through the UKB data-access protocol.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes are clearly indicated in the manuscript. Sample sizes were determined as the total number of available individuals for a specific population with WGS and phenotypic data. Downsampling experieements (roughly multiples of 2) were performed for measuring discovery power.
Data exclusions	We only included samples from the population grouping with the highest sample size, to reduce spurious genetic associations.
Replication	Research findings were replicated through downsampling/out-of-sample experiments. All the code ans scripts used in our experiments are available in a open-source format and can be accessed through our GitHub repository. We have also documented the versions of the software and the tools used to generate the results, enabling other researchers to replicate our experiments with ease.
Randomization	Randomised sample allocation to experimental groups was not performed, as all samples were used together as a group. Out-of-sample validation experiments selected patients completely at random.
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